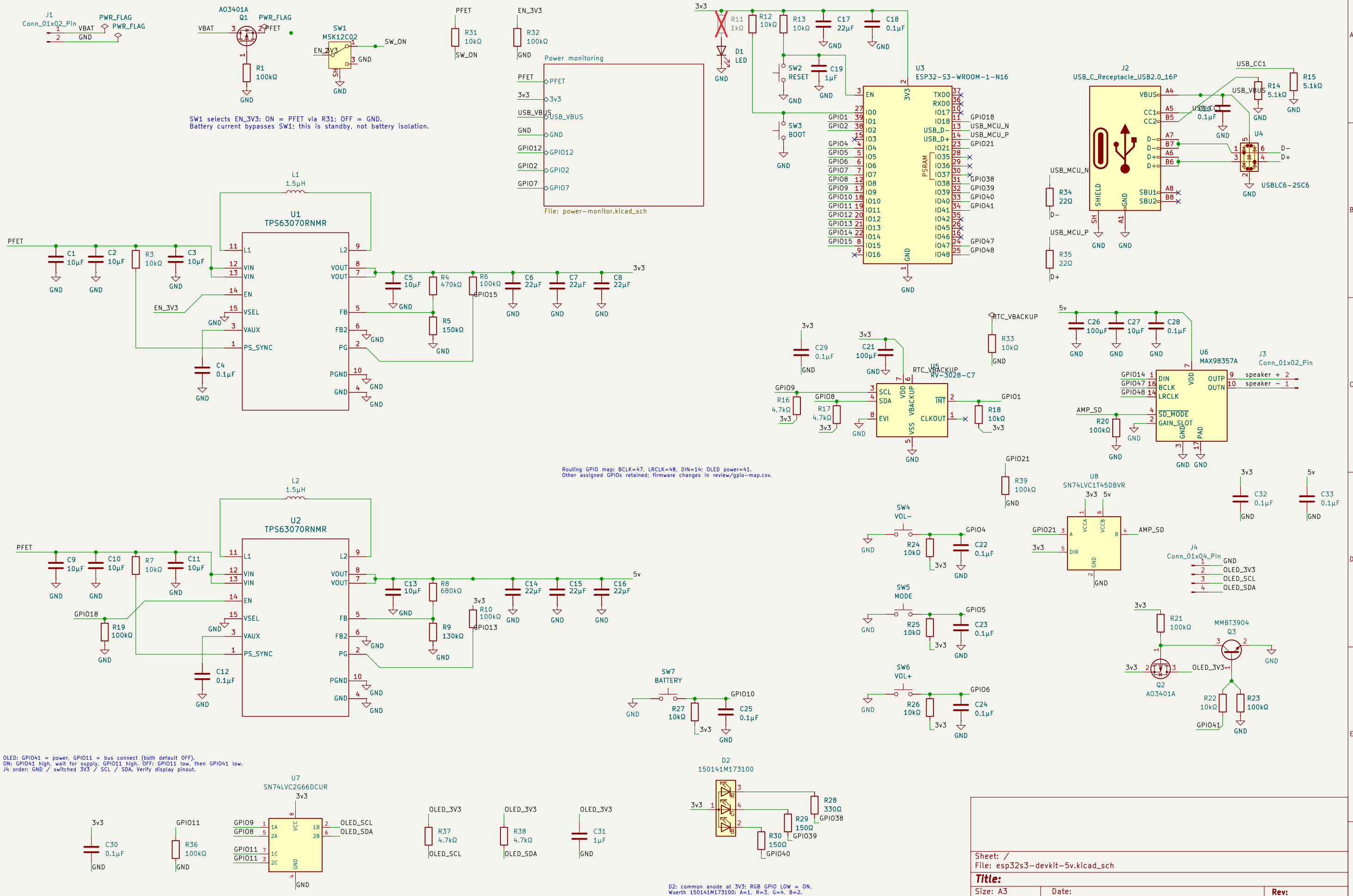


DESIGN REVIEW 2026-09-04 | Prototype; battery/load and enclosure pending confirmation.
TPS63070: PS/SYNC HIGH = power save (both rails); 3.31V / 4.985V nominal.
Cold start requires >=3.0V at converter input. J5-PH Input rated 2A with specified wire.
USB is data-only; VBUS supplies ESD bias only. R11 DNP for low standby current.
RTC backup/trickle charge OFF; time is lost when main power is off.
Capacitance must be checked after DC bias; inspect review/design-review.md before fabrication.



OLED: GPIO41 = power, GPIO11 = bus connect (both default OFF).
ON: GPIO41 high, wait for supply, GPIO11 high, OFF: GPIO11 low, then GPIO41 low.
J4 order: GND / switched 3V3 / SCL / SDA. Verify display pinout.

D2: common anode at 3V3; RGB GPIO LOW = ON.
Wuerth 150141M173100: A=1, R=3, G=4, B=2.

Sheet: /
File: esp32s3-devkit-5v.kicad_sch

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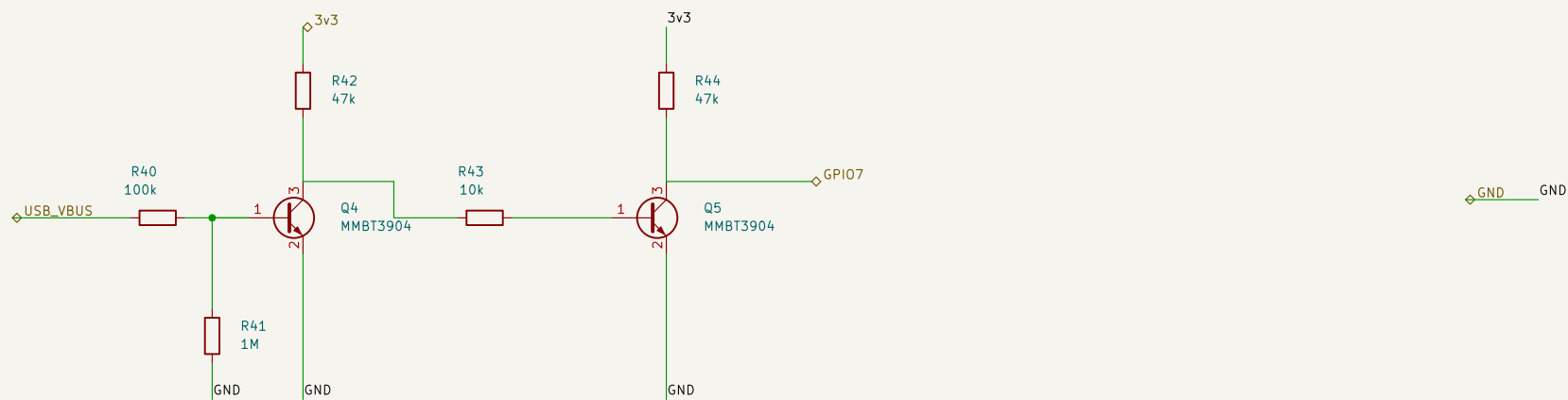
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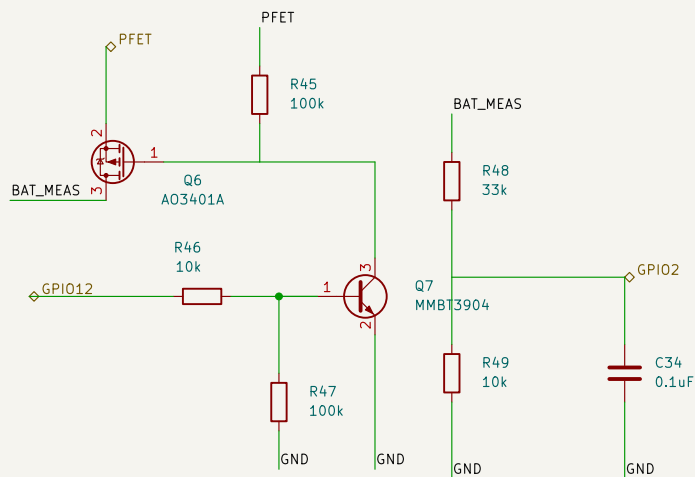
USB VBUS presence – HIGH only with a host supply connected

Two NPN stages keep VBUS away from the GPIO and preserve positive logic.
GPIO7: TinyUSB self_powered=true, vbus_monitor_io=7.
This is cable presence detection, not a precision USB undervoltage comparator.



Switched battery measurement – protected battery node to ADC1 / GPIO2

GPIO12 HIGH enables measurement; wait 10 ms, read calibrated ADC, then drive LOW.
Scale = 4.3; 33k / 10k, 0.1%; 100 nF filter, time constant 0.77 ms.
Designed measurement range 0-6 V; battery chemistry and percentage thresholds remain firmware settings.



Sheet: /Power monitoring/
File: power-monitor.kicad_sch

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